

Tunesh Verma

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AI/ML engineer with 7+ years of experience, currently building low-latency, multi-agent conversational AI on the OpenAI Realtime API for an enterprise sales-training platform, with a background in generative AI/LLM agent systems and production computer vision pipelines.

Skills

Conversational & GenAI: OpenAI Realtime API, Voice AI, Multi-Agent Systems, Agentic AI, Retrieval-Augmented Generation (RAG), Re-ranking, Vector Embeddings, LangChain, LangGraph, HuggingFace, Large Language Models (LLM), LLM Fine-tuning, Prompt Engineering, Generative AI, Natural Language Processing (NLP)

Languages: Python, GoLang, TypeScript, C++, SQL

ML / DL / CV: PyTorch, TensorFlow, Keras, Convolutional Neural Networks (CNN), YOLO, OpenCV, MediaPipe, Computer Vision, Deep Learning, Machine Learning

Cloud & MLOps: AWS, Azure, GCP, Docker, Kubernetes, Airflow, DeepStream, MLOps

Databases: PostgreSQL | **Tools:** Git, WebSockets

Experience

PraxisPro AI — *Senior AI Researcher (Contractor)*

April 2025 – Present

- Worked on the AI layer of **PraxisPlay**, an AI-powered roleplay platform for pharmaceutical sales training, building a real-time conversational experience on the OpenAI Realtime API paired with an in-house multi-agent system.
- Built persona-based HCP roleplay logic, scaling to 240 personas for one enterprise client and 420 for another.
- Designed a multi-agent architecture — including a dedicated "observer" agent that nudges the conversation agent toward better topic direction — producing more natural topic transitions and dynamic, context-aware conversation closures in place of scripted endings.
- Built the client knowledge-base pipeline, ingesting 20+ source documents per client (some 50+ pages) into 150,000+ Q&A pairs used as grounding context for the Realtime model and as ground truth for report generation.
- Built the automated post-call reporting pipeline generating structured session reports and multi-category conversation scores from transcripts; validated compliance and factual accuracy across 100+ test scenarios.
- Implemented a "call focus" feature to steer roleplay conversations toward a specific configured topic.
- Achieved end-to-end conversational response latency of 500ms–1.6s, supporting ~1,000 users running 160–180 sessions per week.
- Extended the pipeline to **PraxisCertify** for deterministic certification conversations, adding a global summary report across a user's certification history.
- Built **PraxisCoach**, using a static but configurable question flow with reports generated

dynamically based on user responses.

- Contributed to the analytics module, surfacing user score trends and manager-facing insights for coaching and performance decisions.

MindNerves Technology — AI Engineer (Freelance), Feb 2025–Mar 2025: Improved a healthcare app's recommendation system and chat-based content delivery, building a RAG pipeline over content descriptions with a re-ranking layer.

Sasken Technology — Senior AI Engineer, Dec 17, 2024–Feb 5, 2025: Worked on WebSocket-based real-time communication for existing route-planning software in the automotive industry.

Tekonika — AI Engineer (Freelance), Nov 2024–Dec 2024: Built a POC for a RAG-based conversational pipeline with a chat model for food-industry clients.

Webtiga, Bangalore — AI Team Lead

October 2023 – November 2024

- **Create Agent:** Built a Gen AI agentic orchestration platform (planner, router, tools, sub-agents via LangGraph/LangChain) letting users create and manage Gen AI bots and Digital Twins.
- **Create Brain:** Built an Airflow-orchestrated pipeline with automated AWS SageMaker training and deployment jobs, letting clients fine-tune LLMs (Mistral, Meta Llama 3) from raw PDF/MP3/MP4 data in ~40 hours per model; delivered 3 fine-tuned models across 2 enterprise clients, with resulting models mintable as NFTs for use within Create Agent.
- Delivered client AI products: **MaiSupport** (AI recovery-counseling bot), **TATA AIA** (RAG-based insurance support platform), **Optimis** (healthcare consultation app).

Vision Group, Bangalore — Machine Learning Engineer

July 2020 – September 2023

- **Autonomous Retail:** Built an autonomous vending cooler (98%+ detection accuracy, 15 SKUs) using DeepStream-based on-device video processing on NVIDIA edge hardware; deployed via OpenVINO on NUC devices with Microsoft, integrating Sony cameras and Custom Vision.
- **Planogram:** Built a DPT/U-Net pipeline to remove background products from shelf images; also built POCs for a Llama 2 retail chatbot and 360°-to-3D product image generation.

Freelancer, Bangalore — Computer Vision Engineer, Feb 2020–Jun 2020: Built a push-up counter (MediaPipe pose estimation) and a brake-pad inspection system (hole counting, diameter/edge measurement via template matching and custom OpenCV edge-detection).

MintM, Bangalore — Computer Vision Engineer, Mar 2019–Jan 2020: Built a multi-camera retail people-counting system (YOLOv3/Keras, centroid tracking, heatmaps, 91%-accurate gender classification) and an end-to-end Return-to-Vendor barcode-detection pipeline for Big Basket.

Synergy Labs, Gurugram — Deep Learning Research Intern, Jun 2018–Dec 2018: Built a one-shot-learning facial recognition system and a table-detection model to crop and extract data from document images.

Education

Ajay Kumar Garg Engineering College, Delhi — B.Tech, Mechanical Engineering — 2014–2018